11.2 Normal Modes

WORKSHEET

The question

STEP III 2014, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s3-q10

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself With X at distance $a + x$ from P and Y at distance $a + y$ from Q, how long is the middle spring XY? Which springs are stretched and which compressed when x and y are positive?

STAGE 1 The equations of motion

Covers: the equations of motion

Ask yourself With $XP = a + x$ and $YQ = a + y$, how long is XY? Which springs are stretched and which compressed? Find the force from each spring on X, taking care with directions. Then use symmetry for Y.

STAGE 2 Decoupling

Covers: the combinations $x - y$ and $x + y$

Pause and try Each equation involves both x and y. Subtract them, then add them. What equation does each of $x - y$ and $x + y$ satisfy?

STAGE 3 The initial conditions

Covers: the combinations $x - y$ and $x + y$

Ask yourself What are x and y at the start, and are the particles moving? Use these to fix the constants in your two solutions.

STAGE 4 Never returning

Covers: Y never returns

Pause and try Write y in terms of t. For Y to be back where it started, what must both cosines equal? What would that say about $\sqrt{3}$?

What's really going on

Normal modes are eigenvectors

Write the equations as a matrix equation:

$$m\begin{pmatrix}\ddot{x}\\ \ddot{y}\end{pmatrix} = -\frac{\lambda}{a}\begin{pmatrix}2 & 1\\ 1 & 2\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}.$$

The combinations that decouple are the eigenvectors of the matrix: $(1,-1)$ with eigenvalue 1, and $(1,1)$ with eigenvalue 3 (Section 7.2). In the first mode, $x = -y$: both particles move the same way along the line (remember that x and y are measured from opposite ends), and the middle spring keeps its length. In the second, $x = y$: they move in opposite directions, squeezing and stretching the middle spring, which makes the frequency higher. The squares of the frequencies are the eigenvalues, multiplied by $\frac{\lambda}{ma}$.

This works for any number of masses and springs. A system of N coupled oscillators has N normal modes, given by the eigenvectors of a symmetric matrix, and the spectral theorem guarantees that they always exist and are perpendicular. Molecules vibrate in normal modes, which is how chemists identify them by the infrared light they absorb. Buildings and bridges have normal modes, which engineers must keep away from the frequencies of wind and earthquakes. And as the number of masses tends to infinity, the normal modes become the harmonics of a vibrating string, the starting point of Fourier analysis.

Coming close without arriving

Although Y never returns exactly, it comes as close as you like. The reason is a theorem of Kronecker: if α is irrational, the multiples $\alpha, 2\alpha, 3\alpha, \ldots$, taken modulo 1, come arbitrarily close to every number in $[0,1]$. So we can find times when ωt is a multiple of 2π and $\sqrt{3}\,\omega t$ is within any tolerance of one. The motion is *quasi-periodic*:

it winds round a torus (one angle for each mode) without ever closing up. The same distinction, between rational and irrational frequency ratios, is at the heart of the KAM theorem on the stability of the solar system.

Try this next

1 **Energy.** Show that the total energy $\frac{1}{2}m(\dot{x}^2+\dot{y}^2)+\frac{\lambda}{2a}\left(x^2+y^2+(x+y)^2\right)$ is constant, and that it splits into a sum of two terms, one for each normal mode, each of which is separately constant.

2 **Pure modes.** How should the particles be started so that the motion is periodic, with period $\frac{2\pi}{\omega}$? With period $\frac{2\pi}{\sqrt{3}\,\omega}$?

3 **A stiffer middle spring.** If the middle spring has modulus μ instead of λ, find the two normal-mode frequencies. For which values of μ/λ is every motion periodic?

4 **Three masses.** Three equal masses are connected in a line by four identical springs between two fixed points. Find the matrix and its eigenvalues $(2-\sqrt{2},\,2,\,2+\sqrt{2})$, and describe the three normal modes.

Further reading

- **A. P. French, *Vibrations and Waves*** (MIT Introductory Physics Series), Chapter 5, on coupled oscillators and normal modes.
- **Gilbert Strang, *Introduction to Linear Algebra*** (Wellesley-Cambridge), the chapter on eigenvalues, which includes springs and masses as a main example.
- **Steven Strogatz, *Nonlinear Dynamics and Chaos*** (CRC Press), Chapter 8, on quasi-periodic motion on a torus.

The full guided solution is in Section 11.2:
https://beyondstep.keithmcnulty.org/ch11/normal-modes.html